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Department of Information Technology

A.Y. 2025-2026
Class: BE-IT A/B, Semester: VIII
Subject: Cloud Computing Lab

Student Name: Student Roll No:

Experiment – 9: Implement Containerization

Aim: To study and Implement Containerization using Docker on AWS/Azure/Google cloud platform.(Free Tier Cloud Platforms).

Objective: After performing the experiment, the students will be able to learn –
- How to implement containerization using docker

Lab objective mapped : ITDO8024.5 : To comprehend Docker and its role in the DevOps lifecycle.

Prerequisite: An AWS account

Requirements: Cloud Login**,** Desktop, Browser, Internet

Pre-Experiment Theory: (describe following topics in soft copy form)
1. **Amazon Elastic Container Service (Amazon ECS)**
2. **Docker applications**
3. **Amazon ECS task definitions**
4. **Application Load Balancer**

Procedure:
https://docs.aws.amazon.com/hands-on/latest/deploy-docker-containers/deploy-docker-containers.html

Post-Experiments Exercise

Deploy Docker Containers on Amazon ECS (YouTube Video):
https://www.youtube.com/watch?v=9hCZYBs0h9I
Watch video

Extended Theory: (to be written in hand)

1. How is Amazon ECS different from AWS Elastic Beanstalk?

Results/Calculations/Observations:
Fill the following observation tables-(to be written in hand)
Amazon Aurora Vs RDS

Sr. No	**Questions**	**Yes/No**	**Justify**
1.	Does Amazon ECS support any other container types?		
2.	Does Amazon ECS support batch jobs?		
3.	Can user use their own Amazon Machine Image (AMI) on ECS?		

Conclusion:
1. Write what was performed in the experiment
2. Mention a few applications of what was studied.
3. Write the significance of the studied topic

References:
[1] [Online]https://ahireharshal.com/cloud/aws-ecs/
[2] [Online]https://docs.docker.com/cloud/ecs-integration/